BRAINSTORM GROUP (BSG)

MAT101
General Mathematics I
Compiled E-Test Questions [2025]

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BRAINSTORM GROUP (BSG) MAT101 E-TEST (2025)

1. Given that $f(x) = \sqrt{x+4}$, compute $f(0)$
   A. 0 B. -1
   C. 1 D. 2

2. Given that $f(x) = \sqrt{x+4}$, compute $f(-2)$
   A. -2 B. $-2i$
   C. $\sqrt{2}$ D. 2

3. Given that $f(x) = \sqrt{x+4}$, compute $f(-5)$
   A. i B. $-i$
   C. 1 D. -1

4. Given that $f(x) = \sqrt{x+4}$, compute $f(-3)$
   A. i B. $-i$
   C. 1 D. -1

5. If $f(x) = x+3, g(x) = x^2+1$ and $h(x) = x$, find gof
   A. x^2+6x+9 B. $x^2+6x+10$
   C. x^2+6x-9 D. $x^2+6x-10$

6. If $f(x) = x+3, g(x) = x^2+1$ and $h(x) = x$, find goh
   A. x^2+1 B. x^2+4
   C. $x+3$ D. x^2+3

7. If $f(x) = x+3, g(x) = x^2+1$ and $h(x) = x$, find fog
   A. x^2+1 B. x^2+4
   C. $x+3$ D. x^2+3

8. If $f(x) = x+3, g(x) = x^2+1$ and $h(x) = x$, find foh
   A. x^2+1 B. x^2+4
   C. $x+3$ D. x^2+3

9. If $f(x) = x+3, g(x) = x^2+1$ and $h(x) = x$, find hof
   A. x^2+1 B. x^2+4
   C. $x+3$ D. x^2+3

10. If $f(x) = x+3, g(x) = x^2+1$ and $h(x) = x$, find hog
    A. x^2+1 B. x^2+4
    C. $x+3$ D. x^2+3

11. If $f(x) = x+3, g(x) = x^2+1$ and $h(x) = x$, find $fogoh$
    A. x^2+1 B. x^2+4
    C. $x+3$ D. x^2+3

12. If $z_1 = 2+3i, z_2 = 4+5i$, find z_1+z_2
    A. $6+8i$ B. $8+6i$
    C. $-6-8i$ D. $-8+6i$

13. If $z_1 = 2+3i, z_2 = 4+5i$, find z_2-z_1
    A. $-2(1+i)$ B. $2(1+i)$
    C. $2(1-i)$ D. $-2(1-i)$

14. If $z_1 = 3+2i, z_2 = 4-5i$, find z_1z_2
    A. $22+7i$ B. $22-7i$
    C. $-22+7i$ D. $-22-7i$

15. Given that $A = 3+i$ and $B = 4+2i$, find AB
    A. $-101-10i$ B. $10+10i$
    C. $10-10i$ D. $-10+10i$

16. Given that $A = 3+i$ and $B = 4+2i$, find $A+B$
    A. $7+3i$ B. $-7+3i$
    C. $7-3i$ D. $-7-3i$

17. Given that $A = 3+i$ and $B = 4+2i$, find $B-A$
    A. $-1-i$ B. $1+i$
    C. $1-i$ D. $-1+i$

18. Calculate (a) the sum of the root and (b) the product of the root of the quadratic equation $x^2-4x+1=0$
    A. $-4, 1$ B. $4, 1$
    C. $4, -1$ D. $-4, -1$

19. Calculate (a) the sum of the root and (b) the product of the root of the quadratic equation $x^2-8x+10=0$
    A. $-8, 10$ B. $8, 10$
    C. $8, -10$ D. $-8, -10$

20. Examine the nature of the root of the following quadratic equation $4x^2+20x+25=0$
    A. distinct real roots
    B. double roots
    C. complex roots
    D. one real and one complex roots

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21. Calculate the sum of the root $2x^2 + 3x = 2$

A. $\frac{3}{2}$ B. $-\frac{3}{2}$

C. 1 D. -1

22. Find the discriminant of the $x^2 + 5x + 6 = 0$

A. 2 B. 25

C. 1 D. -1

23. Find the discriminant of the $3x^2 - 4x + 12 = 0$

A. 72 B. -72

C. 128 D. -128

24. Find the discriminant of the $x^2 + 4x = 4$

A. 0 B. 16

C. 32 D. -16

25. Find the discriminant of the $x^2 + 3x - 1 = 0$

A. $\sqrt{13}$ B. -13

C. 13 D. $-\sqrt{13}$

26. Solve for M given $logM = \frac{1}{4}log81$

A. 4 B. 9

C. 1 D. 3

27. Solve for x given $log_8 x = -\frac{4}{3}$

A. $\frac{1}{16}$ B. $-\frac{1}{16}$

C. 16 D. -16

28. Given $B = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 4 & 3 \\ 5 & 7 & 1 \end{bmatrix}$, compute B^T

A. $\begin{bmatrix} 1 & 5 & 2 \\ 3 & 7 & 4 \\ 4 & 1 & 3 \end{bmatrix}$ B. $\begin{bmatrix} 5 & 2 & 1 \\ 7 & 4 & 3 \\ 1 & 3 & 3 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 2 & 5 \\ 3 & 4 & 7 \\ 4 & 3 & 1 \end{bmatrix}$ D. $\begin{bmatrix} 1 & 3 & 4 \\ 2 & 4 & 3 \\ 5 & 7 & 1 \end{bmatrix}$

29. Given $A = \begin{bmatrix} 2 & 3 & -1 \\ 5 & 1 & 0 \end{bmatrix}$, find A^T

A. $\begin{bmatrix} 2 & 3 & -1 \\ 5 & 1 & 0 \end{bmatrix}$ B. $\begin{bmatrix} 5 & 1 & 0 \\ 2 & 3 & -1 \end{bmatrix}$

C. $\begin{bmatrix} 5 & 2 \\ 1 & 3 \\ 0 & -1 \end{bmatrix}$ D. $\begin{bmatrix} 2 & 5 \\ 3 & 1 \\ -1 & 0 \end{bmatrix}$

30. Find the determinant of the matrix $\begin{bmatrix} 6 & 5 \\ 2 & 3 \end{bmatrix}$

A. 0 B. -8

C. 1 D. 8

31. Given $A = \begin{bmatrix} 2 & 3 \\ 5 & 1 \end{bmatrix}$, find A^T

A. $\begin{bmatrix} 2 & 5 \\ 3 & 1 \end{bmatrix}$ B. $\begin{bmatrix} 3 & 1 \\ 2 & 5 \end{bmatrix}$

C. $\begin{bmatrix} 3 & 2 \\ 1 & 5 \end{bmatrix}$ D. $\begin{bmatrix} 2 & 3 \\ 5 & 1 \end{bmatrix}$

32. Given $B = \begin{bmatrix} 3 & 4 \\ 2 & 3 \\ 5 & 4 \end{bmatrix}$, compute B^T

A. $\begin{bmatrix} 3 & 2 & 5 \\ 4 & 3 & 4 \end{bmatrix}$ B. $\begin{bmatrix} 4 & 3 & 4 \\ 3 & 2 & 5 \end{bmatrix}$

C. $\begin{bmatrix} 4 & 3 \\ 3 & 2 \\ 4 & 5 \end{bmatrix}$ D. $\begin{bmatrix} 3 & 4 \\ 2 & 3 \\ 5 & 4 \end{bmatrix}$

33. The following rules holds for matrix multiplication and addition except

A. $X \cdot Y \neq Y \cdot X$ B. $A \cdot B = B \cdot A$

C. $A + B = B + A$ D. $X + Y = Y \cdot X$

34. Express $\frac{5}{3+\sqrt{2}} - \frac{2}{3-\sqrt{2}}$ in the form $a + b\sqrt{n}$

A. $\frac{9}{7} + 7\sqrt{2}$ B. $\frac{9}{7} - \frac{\sqrt{2}}{7}$

C. $\frac{9}{7} - 7\sqrt{2}$ D. $\frac{9}{7} - \sqrt{2}$

35. Solve the equation $\sqrt{x} - \sqrt{x-2} = 1$

A. $\frac{3}{2}$ B. $\frac{9}{4}$

C. $-\frac{3}{2}$ D. $-\frac{9}{4}$

36. Solve the equation $\sqrt{x+4} - \sqrt{x+5} = 1$

A. 0 B. -2

C. 4 D. -4

37. Evaluate $125^{\frac{2}{3}}$

A. 5 B. 25

C. $\frac{1}{5}$ D. $\frac{1}{25}$

38. Evaluate $32^{\frac{2}{5}} \times 64^{\frac{1}{18}}$

A. $\sqrt[3]{64}$ B. $\sqrt[3]{128}$

C. $\sqrt[3]{256}$ D. $\sqrt[3]{512}$

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39. Simplify $\frac{3\sqrt{2}-\sqrt{3}}{3\sqrt{3}-\sqrt{2}}$

A. $\frac{8\sqrt{6}-3}{5}$ B. $\frac{4\sqrt{6}-6}{5}$

C. $\frac{8\sqrt{6}-3}{25}$ D. $\frac{4\sqrt{6}-6}{25}$

40. Simplify $\sqrt{27}+\sqrt{2}-\sqrt{48}$

A. $\sqrt{2}-\sqrt{3}$ B. $\sqrt{2}+\sqrt{3}$

C. $\sqrt{3}-\sqrt{2}$ D. $-\sqrt{2}-\sqrt{3}$

41. Simplify $3\sqrt{2}-\sqrt{32}+\sqrt{60}$

A. $2\sqrt{15}-\sqrt{2}$ B. $\sqrt{5}-\sqrt{2}$

C. $5\sqrt{2}-\sqrt{15}$ D. $\sqrt{15}-\sqrt{2}$

42. Obtain the equation whose roots are -1 and 5

A. $x^2-4x-5=0$

B. $x^2-5x-4=0$

C. $x^2+4x-5=0$

D. $x^2+5x-4=0$

43. Obtain the roots of quadratic equation $x^2-x=0$

A. $-1, 1$ B. $0, 1$

C. $0, -1$ D. $-2, -1$

44. Evaluate $x-3>1$

A. $x<4$ B. $x>4$

C. $x>-4$ D. $x<-4$

45. Evaluate $2x-3<x+7$

A. $x<10$ B. $x>10$

C. $x>5$ D. $x<5$

46. Evaluate $|x-3|>1$

A. $x<2$ B. $x>4$

C. $x>2$ D. A & B

47. Evaluate $|2x-5|<3$

A. $x>4$ B. $1<x<4$

C. $x>1$ D. A & B

48. Evaluate $|2-4x|\geq 6$

A. $x\geq 2$ B. $x\leq -1$

C. $2\leq x>-1$ D. A & B

49. Obtain the solution of the equation

$$log_x 3+log_x 9+log_x 729=9$$

A. 0 B. 3

C. 6 D. 9

50. Evaluate $\left(\frac{81y^8}{16x^4}\right)^{\frac{3}{4}}$

A. $\frac{3y}{2x}$ B. $\frac{9y^4}{4x^2}$

C. $\frac{27y^6}{8x^3}$ D. $\frac{16y^6}{27x^3}$

51. Given the universal set $\mathcal{U}=\{1,2,3,4,5\}$, $P=\{1,2,4\}$ and $Q=\{2,4,5\}$. Find $P'\cap Q$

A. $\{2,4\}$ B. $\{1,2\}$

C. $\{5\}$ D. $\{4,5\}$

52. Given the universal set $\mathcal{U}=\{1,2,3,4,5,6\}$, $X=\{2,4,6,\}$ and $Y=\{1,2,6\}$. Find $(X\cap Y)^c$

A. $\{2,4\}$ B. $\{1,2\}$

C. $\{1,3,4,5\}$ D. $\{4,5\}$

53. In a class of 100 students, 40 offer chemistry, 35 offer geography, 30 offer neither. How many students of both subject?

A. 4 B. 5

C. 6 D. 9

54. In an examination 18 students passed MAT111, 17 passed PHY113, 11 students passed both. How many passed MAT111 only?

A. 1 B. 3

C. 7 D. 9

55. Express $cos^2\theta$ in terms of multiple angles.

A. $2^{-1}(cos2\theta-1)$ B. $-2^{-1}(cos2\theta-1)$

C. $2^{-1}(cos2\theta+1)$ D. $-2^{-1}(cos2\theta+1)$

56. Determine the zero of the function $2x-3$

A. $\frac{3}{2}$ B. $-\frac{3}{2}$

C. $\frac{3}{2}$ D. $-\frac{3}{2}$

57. Find the cardinality of the set $\{x:-3<x\leq 2\}$

A. 4 B. 5

C. 6 D. 7

58. Find the range of the function $y=7x+2$

A. $(0,\infty)$ B. $(-\infty,0)$

C. $(-\infty,\infty)$ D. $(1,\infty)$

59. Find the range of the function $y=\sqrt{3-x}$

A. $(0,\infty)$ B. $(-\infty,0)$

C. $(-\infty,\infty)$ D. $(-1,\infty)$

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60. Given that $A = 3 + i$ and $B = 4 + 2i$, find BA

A. $-10 - 10i$ B. $10 + 10i$

C. $10 - 10i$ D. $-10 + 10i$

61. What is the value of a given $a + bi = \frac{1-i}{2+i}$

A. $\frac{1}{5}$ B. $-\frac{1}{5}$

C. $\frac{3}{5}$ D. $-\frac{3}{5}$

62. What is the value of b given $a + bi = \frac{1-i}{2+i}$

A. $\frac{1}{5}$ B. $-\frac{1}{5}$

C. $\frac{3}{5}$ D. $-\frac{3}{5}$

63. Find $\frac{z_1}{z_2}$ if $z_1 = 2 - 4i$ and $z_2 = 6 + 7i$

A. $\frac{16+38i}{85}$ B. $\frac{-16+38i}{85}$

C. $\frac{16-38i}{85}$ D. $\frac{-16-38i}{85}$

64. Find the real part of $\frac{2+3i}{3+2i}$

A. $\frac{4}{13}$ B. $\frac{5}{13}$

C. $\frac{12}{13}$ D. 1

65. Find the roots of $x^2 + 8x + 16 = 0$

A. $-2, -4$ B. $-4, -4$

C. $2, -4$ D. $4, -4$

66. Find the roots of $x^2 + 4x - 5 = 0$

A. $-1, -5$ B. $1, -5$

C. $1, 5$ D. $-1, 5$

67. If set $F = (2 < x \leq 5) \cup (9 < x < 12)$, list the members of F.

A. $\{\ \}$

B. $\{3, 4, 5, 10, 11\}$

C. $\{3, 4, 5\}$

D. $\{2, 3, 4, 5, 10, 11\}$

68. Find the range of the function $y = modulus\ of\ 5$

A. $[-5, \infty]$ B. $[-\infty, 5]$

C. $[-5,5]$ D. $[5,5]$

69. Solve the equation:

$log_3 3 + log_3(x + 1) - log_3(2x - 7) = 4$

70. Evaluate $\log_2 16 - 5\log_2 8 - 3\log_4 32$

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ANSWERS

1.	D	11.	B	21.	B	31.	A	41.	A	51.	C	61.	A
2.	C	12.	A	22.	C	32.	A	42.	A	52.	C	62.	D
3.	A	13.	B	23.	D	33.	B&D	43.	B	53.	B	63.	D
4.	C	14.	B	24.	C	34.	D	44.	B	54.	C	64.	C
5.	B	15.	B	25.	C	35.	B	45.	A	55.	C	65.	B
6.	A	16.	A	26.	D	36.	D	46.	D	56.	A	66.	B
7.	B	17.	B	27.	A	37.	B	47.	B	57.	B	67.	B
8.	C	18.	B	28.	C	38.	B	48.	D	58.	C	68.	D
9.	C	19.	B	29.	D	39.	C	49.	B	59.	A	69.	$\frac{190}{53}$
10.	A	20.	B	30.	D	40.	A	50.	C	60.	B	70.	$-\frac{37}{2}$

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